

The reliability and validity of the Polish version of the Breastfeeding Self-Efficacy Scale-Short Form: Translation and psychometric assessment

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Abstract

Background: The majority of women discontinue breastfeeding before the recommended 6 months postpartum. If health professionals are to improve low breastfeeding duration and exclusivity rates, they need to reliably assess high-risk women and identify predisposing factors that are amenable to intervention. One possible modifiable variable is breastfeeding confidence. The Breastfeeding Self-Efficacy Scale Short Form (BSES-SF) is a 14-item measure designed to assess a mother's confidence in her ability to breastfeed her baby.

Objectives: To translate the BSES-SF into Polish and assess its psychometric properties among breastfeeding mothers.
Design: Methodological study.

Setting: Five hospitals in Lodz Poland.

Participants: Using convenience sampling, 105 in-hospital breastfeeding mothers were recruited.

Methods: Following back-translation procedures, mothers completed questionnaires in-hospital and at 8 and 16 weeks postpartum.

Results: The Cronbach's α coefficient for internal consistency was 0.89. In-hospital BSES-SF scores significantly predicted breastfeeding duration and exclusivity at 8 and 16 weeks postpartum. Demographic response patterns suggested the BSES-SF is a unique tool to identify breastfeeding mothers at risk of prematurely discontinuing.

Conclusions: This study provides evidence that the translated version of the BSES-SF may be a valid and reliable measure of breastfeeding self-efficacy among a sample of mothers in Lodz Poland.

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Keywords: Breastfeeding Self-Efficacy Scale; Breastfeeding confidence; Translation; Psychometric assessment; Polish

What is already known about the topic?

The Breastfeeding Self-Efficacy Scale-Short Form (BSES-SF) is a fast, accurate, and systematic assessment tool to identify breastfeeding mothers in the immediate postpartum period who require additional support. It can be used by English researchers and health professionals to (1) identify breastfeeding mothers at high risk

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to prematurely discontinue, (2) assess breastfeeding behaviours and cognitions to individualize confidence-building strategies, and (3) evaluate the effectiveness of various interventions.

What this paper adds

The results from this psychometric investigation suggest that the translated version of the BSES-SF may be a valid and reliable measure of breastfeeding confidence among Polish mothers in Lodz and supports the growing body of literature identifying breastfeeding self-efficacy as an important variable affecting breastfeeding outcomes.

1. Introduction

Current World Health Organization/United Nations Children's Fund (WHO/UNICEF) recommendations for optimal infant feeding are exclusive breastfeeding for approximately the first 6 months postpartum, after which complementary food should be introduced gradually, with the continuation of breastfeeding until 2 years or beyond (WHO, 2003). In the past decade, there has been a myriad of research into the factors affecting breastfeeding duration to identify risk factors for failure. Maternal demographics, attitudes and beliefs, and hospital practices have been examined (Dennis, 2002). Despite the significant research on the barriers to breastfeeding and the many efforts to promote and support it, almost every country in the world fails to meet the WHO recommendations for exclusive breastfeeding (WHO, 2001). For example, in Poland the majority of mothers initiate breastfeeding; however, according to WHO only 68% are exclusively breastfeeding at 4 weeks postpartum decreasing to 37% at 12 weeks and 8% at 24 weeks. Among the mothers who do initiate breastfeeding, 64.6% have discontinued within the first 8 weeks postpartum (Mikiel-Kostyra and Mielniczuk, 1996). While there have been increasing efforts in Poland to promote and support breastfeeding, no national goals exist and health professionals try to promote exclusive breastfeeding for up to 6 months in accordance with the WHO recommendations for infant feeding (WHO, 2003).

Many known predictors of premature breastfeeding discontinuation are *nonmodifiable* demographic variables such as maternal age, marital status, educational level, and socioeconomic status (Dennis, 2002). If health professionals are to effectively improve low breastfeeding duration rates, they need to reliably assess high-risk women and identify predisposing factors that are amenable to intervention (Dennis and Faux, 1999). One possible *modifiable* variable is breastfeeding confidence.

To promote the conceptual development of breastfeeding confidence, Bandura's (1977) social cognitive theory and self-efficacy concept may be used. According to Bandura, self-efficacy is an important health-related behaviour predictor. Defined as an individual's confidence in his or her perceived ability to perform a specific task or behaviour (Bandura, 1977), self-efficacy is composed of two parts: (1) outcome expectancy, the belief that a given behaviour will produce a particular outcome; and (2) self-efficacy expectancy, an individual's conviction that one can successfully perform certain tasks or behaviours to produce the desired outcome (Bandura, 1977). These self-efficacy expectancies influence behaviours individuals engage in, how much effort they expend, how long they persist when faced with obstacles, and whether they undertake self-debilitating or self-encouraging cognitions. Incorporating self-efficacy theory, Dennis (1999) developed the breastfeeding self-efficacy concept. Breastfeeding self-efficacy refers to a mother's confidence in her ability to breastfeed her infant. It is an important variable in breastfeeding outcomes as it predicts: (1) whether a mother chooses to breastfeed or not, (2) how much effort she will expend, (3) whether she will have self-enhancing or self-defeating thought patterns, and (4) how she will emotionally respond to breastfeeding difficulties (Dennis, 1999). Breastfeeding self-efficacy is influenced by four main sources of information: (1) performance accomplishments (e.g., past breastfeeding experiences), (2) vicarious experiences (e.g., watching other women breastfeed), (3) verbal persuasion (e.g., encouragement from influential others such as friends, family, and lactation consultants), and (4) physiological responses (e.g., pain, fatigue, stress, anxiety) (Bandura, 1977; Dennis, 1999). As such, health professionals may enhance a mother's breastfeeding confidence by altering these sources of self-efficacy information.

To measure breastfeeding confidence, the Breastfeeding Self-Efficacy Scale (BSES) was developed and psychometrically tested (Dennis and Faux, 1999). Replicating this original research, further methodological studies have been conducted in Canada (Dennis, 2003), Australia (Blyth et al., 2002, 2004; Creedy et al., 2003), China (Dai and Dennis, 2003), and Puerto Rico (Molina Torres et al., 2003). In these studies, BSES scores in the early postpartum period have consistently predicted breastfeeding duration at 4, 6, 8, and 16 weeks postpartum. In addition, a significant relationship has been demonstrated between BSES scores and exclusive breastfeeding (Blyth et al., 2002, 2004; Creedy et al., 2003; Dai and Dennis, 2003; Dennis, 2003; Molina Torres et al., 2003). These studies provide preliminary evidence that the BSES may be an internationally applicable, reliable, and valid measure to assist health professionals in caring for breastfeeding women. In non-English speaking areas, health professionals would

benefit from having a translated version of the scale. The purpose of this methodological study was (1) to translate and psychometrically assess the Breastfeeding Self-Efficacy Scale-Short Form (BSES-SF) among Polish women and (2) to examine the relationship between breastfeeding self-efficacy and maternal demographic variables.

2. Literature review

2.1. Importance of breastfeeding self-efficacy

The saliency of breastfeeding self-efficacy (defined in earlier research as maternal confidence) for breastfeeding outcomes has been demonstrated consistently by diverse researchers (Buxton et al., 1991; Loughlin et al., 1985). In a recent study of 694 pregnant US women who were certified for the Women, Infants, and Children (WIC) program (a supplemental nutrition program for low-income women and children), breastfeeding confidence significantly predicted intention to breastfeed (Mitra et al., 2004). Similarly, in a descriptive study of 198 pregnant US women, O'Campo et al. (1992) examined 11 psychosocial and demographic variables and found breastfeeding confidence to be one of the most significant variables affecting anticipated length of breastfeeding. Breastfeeding confidence postnatally is also of consequence. In a longitudinal study of 64 minority women eligible for a WIC program, mothers who lacked confidence that they would still be breastfeeding at 8 weeks postpartum were significantly more likely to discontinue before 2 weeks (Ertem et al., 2001). Likewise, in a Danish study of 471 mothers, those with moderate-to-low confidence in their ability to breastfeed were almost 2 times more likely to discontinue breastfeeding before 4 months than mothers with high confidence (Kronborg and Vaeth, 2004). Based on qualitative and quantitative data obtained from three questionnaires administered to 159 Australian women, Papinczak and Turner (2000) found mothers who were unable to establish lactation reported significantly lower levels of breastfeeding confidence in comparison to mothers who breastfed for more than 6 months. In a phenomenological study involving in-depth interviews with primiparous women in the UK, deterioration in breastfeeding confidence during the early postpartum period was a major factor in the decision to discontinue breastfeeding (Dykes and Williams, 1999). In another qualitative study that incorporated focus groups and in-depth semi-structured interviews with adolescent mothers, one of the five themes that emerged related to their breastfeeding experience was a lack of confidence in their ability to breastfeed and produce sufficient milk (Dykes et al., 2003). In two other studies,

cient milk supply and breastfeeding confidence was found (Hill and Humenick, 1996; Segura-Millan et al., 1994). This is a noteworthy finding considering that perceived insufficient milk supply is a leading cause of formula supplementation and a decrease in breastfeeding level (Dennis, 2002). These preceding results provide evidence that breastfeeding self-efficacy is a salient variable in the initiation and duration of breastfeeding.

2.2. Development of the Breastfeeding Self-Efficacy Scale-Short Form

The BSES (Dennis and Faux, 1999) is a 33-item, self-report instrument developed to measure breastfeeding confidence. The measure contains two subscales: the technique subscale, where items depict maternal skills and recognition of specific principles required for successful breastfeeding, and the intrapersonal thoughts subscale, where items are related to maternal attitudes and beliefs towards breastfeeding. All items are preceded by the phrase "I can always" and anchored with a five-point Likert scale where one is not at all confident and five is always confident. As recommended by Bandura (1977), all items are presented positively and scores are summed to produce a range from 33 to 165 with higher scores indicating higher levels of breastfeeding self-efficacy. Content validity of the BSES was based on a literature review, interviews with breastfeeding mothers, and expert judgment using a method recommended by Lynn (1986). Following a pilot test, an initial psychometric assessment was conducted with a convenience sample of 130 Canadian breastfeeding women where questionnaires were completed during the postpartum hospitalization and again at 6 weeks postpartum (Dennis and Faux, 1999). Cronbach's α coefficient for the scale was 0.96 with 73% of all corrected item-total correlations ranging from 0.30 to 0.70. Responses were subjected to principal components analysis with a varimax rotation, yielding the theorized subscales. Support for predictive validity was demonstrated through positive correlations between BSES scores and infant feeding method at 6 weeks postpartum.

However, internal consistency statistics with the original BSES suggested item redundancy. As such, another methodological study was undertaken and 18 items were deleted using explicit reduction criteria (Dennis, 2003). Based on the encouraging reliability analysis of the new 14-item BSES-SF, construct validity was assessed using principal components factor analysis, comparison of contrasted groups, and correlations with measures of similar constructs. Support for predictive validity was demonstrated through significant mean differences between breastfeeding and bottle-feeding mothers at 4 and 8 weeks postpartum. Demographic response patterns suggested the BSES-SF is a unique

tool to identify mothers at risk of prematurely discontinuing breastfeeding.

3. Methods

3.1. Design

This methodological study translated and then examined the reliability and validity of the BSES-SF. Participants were recruited in Lodz, one of the largest cities in Poland, from five diverse urban hospitals between June and August 2004. Eligible participants were all in-hospital breastfeeding mothers who were (1) 16 years of age or older (2) able to read and speak Polish, and (3) at least 37 weeks gestation when they gave birth. Mothers were excluded if they had a factor which could significantly interfere with breastfeeding such as multiple or high-risk pregnancy (i.e., serious medical condition or known birth defect). Independent of hospital site, all potential participants were identified on the postpartum ward by the researcher and approached to further assess eligibility and provide study explanation. After informed consent procedures, approved by the Lodz University ethical review board, the translated BSES-SF and demographic questionnaire were completed before hospital discharge. All mothers approached agreed to participate in the study and no incentives were provided. During the in-hospital data collection session, mothers were asked about their current breastfeeding status. For this study, breastfeeding was defined as the receipt by the infant of any breast milk within the past 24 hours and further classified into one of the three following categories: (1) exclusive breastfeeding (e.g., breast milk only and other fluids on a non-daily basis); (2) partial breastfeeding (e.g., at least 1 bottle of formula per day); and (3) bottle-feeding (e.g., no breast milk at all). Mothers who were pumping were considered breastfeeding using the preceding defined categories. No interventions were provided to encourage the continuation of breastfeeding. Data collection procedures were repeated at 8 and 16 weeks via telephone to obtain information regarding infant feeding status.

3.2. Translation of the BSES-SF

To translate the BSES-SF into Polish, diverse methods were used to ensure content, semantic, and technical equivalence was ascertained. The essence of semantic equivalence is that the meaning of each item remains the same after translation into the target language (Flaherty et al., 1988). The most common and highly recommended procedure for establishing semantic equivalence is back-translation (Brislin, 1970; Chapman and Carter, 1979). In addition, some researchers (Butcher, 1982) suggest ‘blind

back-translation’ should be employed such that one back-translates without having knowledge of the original version of the instrument. Five bilingual graduate university students not associated with the project or the breastfeeding field completed the initial translation and five other bilingual graduate university students then back translated the scale into English. The objective in selecting non-professional translators was to guard against language not commonly understood by the average Polish-speaking mother. The five back-translations were judged semantically equivalent to the original BSES-SF by the research team. For cross-cultural research, content equivalence is established by determining whether the content of each item of the instrument is relevant to the target culture (Flaherty et al., 1988). To accomplish this, content experts evaluate the similarity of each item. For this study, 31 mothers who successfully breastfed evaluated the content of each BSES-SF item to determine the relevance to successful breastfeeding and to ensure clarity. Based on this evaluation, two item modifications were made: (1) the item “I can always finish feeding my baby on one breast before switching to the other breast” was changed to “I can always start feeding my baby on the other breast only after emptying the first one” and (2) the item “I can always continue to breastfeed my baby for every feeding” was changed to “I can always ensure that every meal of my baby is breast milk”. The mothers also had difficulty understanding the original response format and changes were made such that 1 is definitely not, 2 is rather not, 3 is uncertain, 4 is rather yes, and 5 is definitely yes. The modified instrument was presented to the breastfeeding mothers a final time and after several discussions all items were considered relevant to Polish-speaking mothers and no further cultural differences were identified.

Pre-testing was completed with 10 additional breastfeeding mothers to evaluate the comprehension and readability of the BSES-SF Polish version. The results showed that all the mothers easily understood the items. As such, the final version of the translated BSES-SF incorporated 14 items with two requiring minor modifications. Finally, to establish technical equivalence, the translated BSES-SF was administered using the paper and pencil method to hospitalized breastfeeding mothers during the first days postpartum, in a manner consistent with the original methodological study (Dennis and Faux, 1999).

4. Results

4.1. Sample characteristics

While 126 breastfeeding women agreed to participate in this study, only 105 ($n = 83.3\%$) responded to the infant feeding status questions at 8 and 16 weeks and

were included in the analysis. This sample size provided more than the minimum requirement of five participants per item needed for psychometric testing (Dennis & Faux, 1999). Mothers who completed all questionnaires did not differ significantly from non-completers ($n = 21$) related to important demographic characteristics including age, education, marital status, and breastfeeding level. The mean age of the sample was 28 years ($SD = 4.5$), ranging from 17 to 42. Six percent ($n = 6$) of mothers had only primary school education while 36.2% ($n = 38$) had secondary education, and 58.1% ($n = 61$) had some postsecondary education. The majority of mothers ($n = 100$, 95.2%) were living with their partner. Seventy percent ($n = 73$) of mothers were primiparous and 44.8% ($n = 47$) had a caesarean delivery. Many of the mothers ($n = 78$, 74.3%) had no previous breastfeeding experience. Of the 25.7% ($n = 27$) of mothers who did previously breastfeed, 5 (4.8%) indicated they had a negative experience. Many of the women reported they had someone close in their lives who had successfully breastfed previously ($n = 87$, 82.9%). Birth and breastfeeding complications were not assessed nor was maternal employment status. It is unknown if the caesarean sections performed were due to maternal choice or medical complications.

In-hospital, 47 (44.8%) mothers were exclusively breastfeeding with 58 (55.2%) partially breastfeeding. At 8 weeks postpartum, 68 (64.8%) mothers were exclusively breastfeeding and 16 (15.2%) were partially breastfeeding; 21 (20%) mothers had discontinued breastfeeding. At 16 weeks postpartum, 60 (57.1%) women were exclusively breastfeeding, 11 (10.5%) were partially breastfeeding, and 34 (32.4%) were bottle-feeding.

4.2. Internal consistency

The reliability of the translated BSES-SF was evaluated using the following criteria: (1) Cronbach's α coefficient, (2) corrected item-total correlation, and (3) α estimate when an item was dropped from the scale. Poorly functioning items were defined as: (1) items that when deleted increased the α coefficient by more than 0.10 or (2) items that had a corrected item-total correlation less than 0.30. This criterion was used in previous BSES psychometric investigations and was used to ensure comparability of results (Dai and Dennis, 2003; Dennis, 2003; Dennis and Faux, 1999; Molina Torres et al., 2003). The Cronbach's α coefficient was 0.89 and was not increased by more than 0.10 if any of the items were deleted. All corrected item-total correlations were positive and above 0.30 with all falling within the recommended range of 0.30–0.70. The translated BSES-SF mean score was 55.5 ± 8.4 , ranging from 34 to 70 with an item mean of 3.97, ranging from 3.23 to 4.31, and an item variance of 0.89, ranging from 0.51 to 1.30.

Finally, the mean inter-item correlation was 0.36 ranging from 0.06 to 0.70.

4.3. Construct validity

According to Bandura (1977), self-efficacy is influenced by previous performance accomplishments. In this study, it was hypothesized that mothers who had previously breastfed would have higher levels of breastfeeding self-efficacy than those with no previous breastfeeding experience. Consistent with this hypothesis, the mean BSES-SF score for women with previous breastfeeding experience was significantly higher (Mean = 60.4, $SD = 6.5$) than BSES-SF scores for women with no previous experience (Mean = 54.2, $SD = 8.4$; $t(103) = -1.39$, $p = 0.002$). In addition, significant differences in BSES-SF mean scores were found between primiparous (Mean = 54.25, $SD = 7.7$) and multiparous (Mean = 58.44, $SD = 9.3$; $t[103] = -2.4$, $p = 0.018$) women. While it was hypothesized that women who vicariously observed successful breastfeeding via their family members and friends would have higher breastfeeding self-efficacy than those who did not, no significant difference between the two groups was found ($t[103] = 0.14$, $p = 0.89$).

4.4. Predictive validity

Predictive validity was determined through the examination of mothers' breastfeeding self-efficacy scores in-hospital and their infant feeding status at 8 and 16 weeks postpartum. Significant differences were found in breastfeeding self-efficacy scores among mothers who at 8 weeks postpartum were either breastfeeding (Mean = 56.7, $SD = 8.1$) or bottle-feeding (Mean = 50.7, $SD = 7.9$; $t[103] = 3.07$, $p = 0.003$). Significant differences were also found among mothers who at 16 weeks postpartum were either breastfeeding (Mean = 57.9, $SD = 7.3$) or bottle-feeding (Mean = 50.6, $SD = 8.5$; $t[103] = 4.53$, $p = 0.001$).

Mothers were further categorized according to their breastfeeding level and a one-way analysis of variance was undertaken. Mothers who were exclusively breastfeeding at 8 weeks postpartum had significantly higher BSES-SF scores (Mean = 58.1, $SD = 7.5$) than mothers who were either partially breastfeeding (Mean = 52.9, $SD = 8.6$) or bottle-feeding (Mean = 50.7, $SD = 7.9$; $F[2] = 10.8$, $p = 0.001$). Similarly, mothers who were exclusively breastfeeding at 16 weeks postpartum had higher BSES-SF scores (Mean = 58.0, $SD = 7.0$) than partially breastfeeding mothers (Mean = 55.7, $SD = 9.4$) or those who were bottle-feeding (Mean = 50.6, $SD = 8.6$; $F[2] = 10.1$, $p = 0.001$). Thus, the higher the BSES-SF score was in-hospital, the more likely the mother was breastfeeding at 8 and 16 weeks postpartum and doing so exclusively.

4.5. Breastfeeding self-efficacy and demographic factors

Differences in breastfeeding self-efficacy were assessed in relation to maternal demographic variables. A weak relationship was found between maternal age and breastfeeding self-efficacy in the immediate postpartum period ($r = 0.21$; $p = 0.03$). No significant differences in BSES-SF scores were found between mothers who had a partner (Mean = 55.2, SD = 8.4) and those who were single (Mean = 61.4, SD = 7.1; $t[103] = 1.61$; $p = 0.11$). In relation to educational level, mothers who had primary school education did not differ significantly in their breastfeeding self-efficacy (Mean = 52.2, SD = 11.9) than mothers with secondary (Mean = 56.2, SD = 7.9) or some postsecondary education (Mean = 55.4, SD = 8.4; $F[2] = 0.61$; $p = 0.54$). Although there was no difference in level of breastfeeding, mothers who had a vaginal birth had lower breastfeeding self-efficacy scores (Mean = 54.0, SD = 8.7) than mothers who gave birth via caesarean section (Mean = 57.4, SD = 7.8; $t(102) = -2.08$, $p = 0.04$).

5. Discussion

The results from this psychometric investigation are consistent with the original study (Dennis, 2003) and provide evidence that the translated version of the BSES-SF may be a valid and reliable measure of breastfeeding confidence among the representative sample of Polish mothers in Lodz. This study also provides further evidence of the international applicability of the BSES-SF and demonstrates that measures can be translated while maintaining reliability and validity. The Cronbach's α coefficient was 0.89, exceeding the recommended α for established instruments (Nunnally and Bernstein, 1994), and is comparable to the original BSES-SF Cronbach's α coefficient of 0.94. The overall mean score of 55.5 is identical to the original BSES-SF mean score of 55.8 (Dennis, 2003). Despite the change in response format, the item mean of 3.97 is consistent with the original BSES-SF item mean of 3.99. The item variance was 0.89, slightly lower than the original 1.04 and the mean inter-item correlation was also slightly lower. Similar to other translated versions of the BSES-SF (Dai and Dennis, 2003; Molina Torres et al., 2003), minor modifications to two items were required for cultural equivalence; a comparison between the translated and original item means did not show a difference (average mean of both items 3.56 versus 3.61 for the original BSES-SF items). As such, the consistency in the item analysis between the original and translated version of the BSES-SF suggests the change in response format and the two items did not substantially alter the reliability of the translated BSES-SF. Providing evidence for construct validity and consistent with

previous research (Dai and Dennis, 2003; Dennis, 2003; Dennis and Faux, 1999; Molina Torres et al., 2003) and breastfeeding self-efficacy theory (Dennis, 1999), primiparous mothers had significantly lower breastfeeding self-efficacy than multiparous women.

The most clear and significant finding reported is the predictive validity of the instrument. Although the measure to operationalize infant feeding status was limited to self-report, the relationship between breastfeeding self-efficacy and breastfeeding duration and exclusivity is consistent with previous research (Dai and Dennis, 2003; Dennis, 2003; Dennis and Faux, 1999; Molina Torres et al., 2003) and is promising for clinical practice. Also important is the lack of association between maternal demographic factors and breastfeeding self-efficacy, suggesting the BSES-SF is a unique tool in identifying mothers at high risk to discontinue breastfeeding prematurely. This finding has been demonstrated with Chinese (Dai and Dennis, 2003) and Australian (Creedy et al., 2003) mothers. However, the finding that mothers who experienced a caesarean delivery had higher breastfeeding self-efficacy than mothers who delivered vaginally conflicts with previous BSES-SF results (Dennis, 2003) and other studies indicating that there may be short-term psychological outcomes related to caesarean childbirth (Clement, 2001; DiMatteo et al., 1996). Additional research is warranted to examine this finding, especially since in this study almost half of the mothers had a caesarean delivery and some researchers suggest rates are increasing internationally (Anderson, 2004).

This study supports the growing body of literature identifying breastfeeding self-efficacy as an important variable affecting breastfeeding outcomes. Based on Bandura's (1977) social cognitive theory and Dennis's (1999) breastfeeding self-efficacy model, a mother's confidence in her ability to breastfeed influences her decision as to whether she will initiate breastfeeding, how long she will continue, and if she will do so exclusively. Using the BSES-SF, health professionals may identify mothers with low breastfeeding self-efficacy that are at risk to prematurely discontinue breastfeeding and implement individualized confidence-enhancing strategies. Given the financial and time constraints currently influencing the provision of health care, it is reasonable to target new mothers with low breastfeeding self-efficacy who are at increased risk to discontinue prematurely.

6. Conclusions

The findings from this study suggest the translated BSES-SF may be a reliable and valid measure of maternal breastfeeding self-efficacy and be predictive of breastfeeding duration and exclusivity at 8 and 16

weeks postpartum. Additional research is warranted to assess breastfeeding behaviours and cognitions to individualize confidence-building strategies and to evaluate the effectiveness of various interventions.

7. Limitations and recommendations

Several study limitations should be highlighted. Due to costs, professional translators were not used in the initial translation of the BSES-SF. The sample was homogeneous in relation to ethnicity and marital status and in combination with the non-random selection process, participants may not have been representative of all Polish women but only those in Lodz. In addition, participants only completed the BSES-SF in-hospital, thus limiting the ability to assess breastfeeding self-efficacy across the postpartum period. Over 82% of mothers in this study had previously observed someone successfully breastfeed. As such, we had insufficient variability to detect differences in breastfeeding self-efficacy between mothers who had and those who had not seen someone successfully breastfeed previously. Due to previous psychometric testing with the BSES-SF suggesting a unifactorial structure (Dennis, 2003), a factor analysis was not completed. However, further evidence for the unifactorial structure is recommended to confirm this original finding. No research to date has been conducted demonstrating that health professionals can actually increase a mother's confidence to breastfeed or whether increased confidence postnatally significantly improves breastfeeding outcomes. It is also unknown which confidence-enhancing strategies are most effective in increasing a mother's breastfeeding self-efficacy nor is it known what time period is most effective time to implement these strategies (e.g., antenatally or postnatally). Additional research is recommended to answer these questions, with the use of the BSES-SF as an evaluation measure.

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References

- Anderson, G., 2004. Making sense of rising caesarean section rates. *British Journal of Medicine* 329, 696–697.
- Bandura, A., 1977. Self-efficacy: toward a unifying theory of behavioral change. *Psychological Review* 84, 191–215.
- Blyth, R., Creedy, D.K., Dennis, C.L., Moyle, W., Pratt, J., De Vries, S.M., 2002. Effect of maternal confidence on breastfeeding duration: an application of breastfeeding self-efficacy theory. *Birth* 29, 278–284.
- Blyth, R.J., Creedy, D.K., Dennis, C.L., Moyle, W., Pratt, J., De Vries, S.M., Healy, G.N., 2004. Breastfeeding duration in an Australian population: the influence of modifiable antenatal factors. *Journal of Human Lactation* 20, 30–38.
- Brislin, R.W., 1970. Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology* 1, 185–216.
- Butcher, J.N., 1982. Cross-cultural research methods in clinical psychology. In: Kendall, P., Butcher, J.N. (Eds.), *Handbook of Research Methods in Clinical Psychology*. Wiley, New York, pp. 273–308.
- Buxton, K.E., Gielen, A.C., Faden, R.R., Brown, C.H., Paige, D.M., Chwalow, A.J., 1991. Women intending to breastfeed: predictors of early infant feeding experiences. *American Journal of Preventive Medicine* 7, 101–106.
- Chapman, D.W., Carter, J.F., 1979. Translation procedures for the cross cultural use of measurement instruments. *Educational Evaluation Policy Analysis* 1, 71–76.
- Clement, S., 2001. Psychological aspects of caesarean section. *Best Practices and Research in Clinical Obstetrics and Gynecology* 15, 109–126.
- Creedy, D.K., Dennis, C.L., Blyth, R., Moyle, W., Pratt, J., De Vries, S.M., 2003. Psychometric characteristics of the breastfeeding self-efficacy scale: data from an Australian sample. *Research in Nursing & Health* 26, 143–152.
- Dai, X., Dennis, C.L., 2003. Translation and validation of the Breastfeeding Self-Efficacy Scale into Chinese. *Journal of Midwifery and Women's Health* 48, 350–356.
- Dennis, C.L., 1999. Theoretical underpinnings of breastfeeding confidence: a self-efficacy framework. *Journal of Human Lactation* 15, 195–201.
- Dennis, C.L., 2002. Breastfeeding initiation and duration: a 1990–2000 literature review. *Journal of Obstetric Gynecologic and Neonatal Nursing* 31, 12–32.
- Dennis, C.L., 2003. The breastfeeding self-efficacy scale: psychometric assessment of the short form. *Journal of Obstetric Gynecologic and Neonatal Nursing* 32, 734–744.
- Dennis, C.L., Faux, S., 1999. Development and psychometric testing of the Breastfeeding Self-Efficacy Scale. *Research in Nursing & Health* 22, 399–409.
- DiMatteo, M.R., Morton, S.C., Lepper, H.S., Damush, T.M., Carney, M.F., Pearson, M., Khan, K.L., 1996. Cesarean childbirth and psychosocial outcomes: a meta-analysis. *Health Psychology* 15, 303–314.
- Dykes, F., Williams, C., 1999. Falling by the wayside: a phenomenological exploration of perceived breast-milk inadequacy in lactating women. *Midwifery* 15, 232–246.
- Dykes, F., Moran, V.H., Burt, S., Edwards, J., 2003. Adolescent mothers and breastfeeding: experiences and support needs—an exploratory study. *Journal of Human Lactation* 19, 391–401.
- Ertem, I.O., Votto, N., Leventhal, J.M., 2001. The timing and predictors of the early termination of breastfeeding. *Pediatrics* 107, 543–548.
- Flaherty, J.A., Gaviria, F.M., Pathak, D., Mitchell, T., Wintrob, R., Richman, J.A., Birz, S., 1988. Developing instruments for cross-cultural psychiatric research. *Journal of Nervous and Mental Disease* 176, 257–263.

- Hill, P.D., Humenick, S.S., 1996. Development of the H & H Lactation Scale. *Nursing Research* 45, 136–140.
- Kronborg, H., Vaeth, M., 2004. The influence of psychosocial factors on the duration of breastfeeding. *Scandinavian Journal of Public Health* 32, 210–216.
- Loughlin, H.H., Clapp-Channing, N.E., Gehlbach, S.H., Pollard, J.C., McCutchen, T.M., 1985. Early termination of breast-feeding: identifying those at risk. *Pediatrics* 75, 508–513.
- Lynn, M.R., 1986. Determination and quantification of content validity. *Nursing Research* 35, 382–385.
- Mikiel-Kostyra, K., Mielniczuk, H., 1996. Karmienie piersią niemowląt i dzieci do lat dwóch. *Pediatrica Polska* 71, 1155–1158.
- Mitra, A.K., Khoury, A.J., Hinton, A.W., Carothers, C., 2004. Predictors of breastfeeding intention among low-income women. *Maternal and Child Health Journal* 8, 65–70.
- Molina Torres, M., Davila Torres, R.R., Parrilla Rodriguez, A.M., Dennis, C.L., 2003. Translation and validation of the breastfeeding self-efficacy scale into Spanish: data from a Puerto Rican population. *Journal of Human Lactation* 19, 35–42.
- Nunnally, J.C., Bernstein, I.H., 1994. *Psychometric Theory*, 3rd ed. McGraw-Hill, New York.
- O'Campo, P., Faden, R.R., Gielen, A.C., Wang, M.C., 1992. Prenatal factors associated with breastfeeding duration: recommendations for prenatal interventions. *Birth* 19, 195–201.
- Papinczak, T.A., Turner, C.T., 2000. An analysis of personal and social factors influencing initiation and duration of breastfeeding in a large Queensland maternity hospital. *Breastfeeding Review* 8, 25–33.
- Segura-Millan, S., Dewey, K.G., Perez-Escamilla, R., 1994. Factors associated with perceived insufficient milk in a low-income urban population in Mexico. *Journal of Nutrition* 124, 202–212.
- WHO, 2001. World Health Organization. The Optimal Duration of Exclusive Breastfeeding. Note for the press no. 7. Available at: <http://www.who.int/inf-pr-2001/en/note2001-07.html>.
- WHO, 2003. The Global Strategy on Infant and Young Child Feeding. A joint WHO/UNICEF statement. World Health Organization, Geneva, Switzerland.